

# Power Consumption Predicting and Anomaly Detection Based on Long Short-Term Memory Neural Network

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**Abstract**—Identification of abnormal power consumption data is a critical and difficult task to guarantee efficiency and reliability of the smart grid. This paper proposes a power consumption predicting and anomaly detection approach based on LSTM (Long Short-Term Memory) neural network. Compared to the ARIMA (Autoregressive Integrated Moving Average) algorithm, LSTM model demonstrates a decrease of 22% in forecasting error. Our simulations are performed based on actual power consumption data of ten thousands of users from Chongqing, China in the period of one year. Different from previous studies, we predict electricity consumption and detect anomaly in long-term prospective. We focus on seasonal and monthly trends, instead of power consumption pattern from typical commercial and residential buildings at metering granularity at once every 15 or 30 minutes. Electricity theft identification is significantly improved compared to previous unsupervised algorithms.

**Keywords**—power consumption prediction; anomaly detection; LSTM; ARIMA

## I. INTRODUCTION

Modern smart grid is an enhanced electrical grid infrastructure with advanced metering and information communication technologies to improve the efficiency, reliability and security of the power grid [1].

Widely deployed Advanced Metering Infrastructure (AMI) and the State Grid Corporation of China (SGCC)'s electric energy data acquisition system (EEDAS) allows real-time collection, transmission and storage of power consumption data [2].

Power consumption forecasting is important for power companies in the factor of power generation scheduling and power dispatching, as well as benefits power consumers who can optimize their usage schedule to achieve a lower cost [3]. Abnormal meter measurements could be caused by the unexpected meter failure or the deliberate meter manipulation. Anomaly detection is an important work to identify energy theft and electricity inefficiencies [4].

According to that intelligent electric meters which supporting metering granularity at once every 15 minutes are not full coverage in china, majority of power consumption data analysis and monitoring still depends on the meter reading data at one point per day. Except for external events, weather information and the calendar information could also have effect on the power consumption curve.

Above considered, this paper proposes a power consumption forecasting model using LSTM. This kind of Recurrent Neural Network (RNN) we build is proved to model the power sophisticated consumption time series as well as other seasonality dependencies about weather and holiday information. It shows outstanding performance results in terms of accuracy by reducing the Root Mean Squared Error (RMSE) by 22% while we compare our LSTM model with ARIMA, another representative algorithm widely used in forecasting signals.

Then, we use the LSTM model to detect anomaly. By calculating if the point value varies from the forecasted enough, we find unexpected drops, spikes and trend changes to deem it as an anomaly.

The analysis process is aimed at using data in last few months or years to find out monthly or annually abnormal behaviors. Compared to the anomaly detection methods designed for daily patterns, our analysis and detection is more practical and accurate.

## II. RELATED WORK

There have been plenty of studies about power consumption forecasting and anomaly detection due to its significance in power industry.

Time-series model like ARIMA is used to capture the time series characteristics of power consumption and predict the forecast future behaviors in [5] [6]. Krishna et al. [7] predict electricity consumption at a half-hour granularity using ARIMA models. ARIMA is based on an approach that several points from the past generate a forecast of the next point with the addition of some random variable, which is usually white noise. What difficult part of this method is selection of the number of differences, number of autoregressions, and error coefficients. For evaluation of our model, we also build the ARIMA model to compare the accuracy and efficiency with.

LSTM has been proved to be able to learn higher-level temporal patterns without prior knowledge of the pattern duration and to detect anomalies [8]. RNN algorithm has demonstrated its validity in fields like speech recognition [9] and language modeling [10]. Cheng et al. [11] explored the effectiveness of LSTM in power demand forecasting area and found that LSTM outperforms both Gradient Boosting Tree (GBT) and Support Vector Regression (SVR) [12]. Liu et al. use deep learning algorithm with the Hidden Markov

Model (HMM) and Semi-Markov Model (SMM) to do the accurately prediction with fluctuations of data considered [13,14]. Considering the factors that have great influence on the fluctuation of power consumption, we also chose the features includes historical consumption data, the weather data and the calendar information. Except for power consumption prediction in [8], we did a step further to detect anomaly based on the LSTM. Janetzko et al. [15] introduce novel clustering-based anomaly detection algorithm and visualize anomaly scores to guide the analyst to abnormal points. According to finding the fraudulent electricity consumers, our LSTM model show more efficiency than unsupervised algorithm.

### III. DATASETS

#### A. Description

The dataset contains power usage data for 21,623 users from one power supply station located in Chongqing, China for the period of nearly one year (2015-12-01~2016-12-31). Along with the power consumption data, weather information including highest temperature and lowest temperature during the day, wind power and direction, weather description. Wind power can be leveled from level 1 to 4. Weather can be classified to 24 types such as sunny, cloudy, light rain, shower, thundershower, heavy snow and so on. A sample is shown in Table I.

#### B. Dataset Preparation

We prepare the input dataset for the LSTM. This involves framing the dataset as a supervised learning problem, transforming time series to stationary by differencing the data and normalizing the input variables. We frame the supervised learning problem as predicting the power consumption data at the current day ( $t$ ) given the historical consumption data and other weather and calendar information at the prior time step. The wind speed, weather description features are one-hot encoded and all features are normalized. NaN (not a number) values were filled with average value during the nearest 7 days.

The trend of some normalized features is shown in Fig. 1.

### IV. METHODOLOGY

#### A. Long Short-Term Memory Model

TABLE I. DATA SAMPLE

| User id              | date           | Meter reading (kWh) | Highest temperature (°C) | Lowest temperature(°C)         |
|----------------------|----------------|---------------------|--------------------------|--------------------------------|
| 1                    | 2016-12-01     | 10753.92            | 16                       | 11                             |
| Weather Description  | Wind direction | Wind power          | Day of week              | If vacation( includes weekend) |
| Cloudy to light rain | Breeze         | Level 3             | 4                        | 0                              |

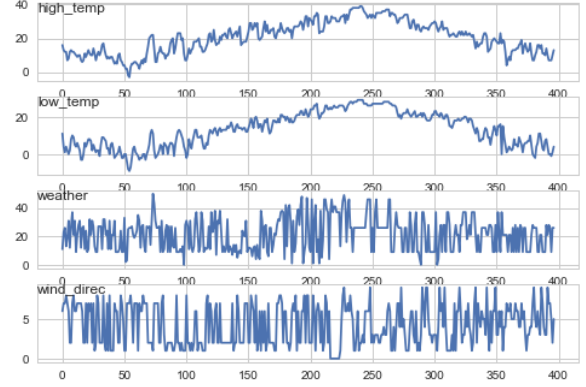


Figure 1. Normalized weather features.

The LSTM network is a recurrent neural network that is trained using *Back propagation Through Time* and overcomes the vanishing gradient problem. LSTM network has memory blocks that are connected through layers.

A block contains gates that manage the block's state and output. There are three types of gates within a unit. First is *Forget Gate*, which conditionally decides what information to throw away from the block. Second is *Input Gate* which conditionally decides which values from the input to update the memory state. Third is *Output Gate* which conditionally decides what to output based on input and the memory of the block. Each unit is like a mini-state machine where the gates of the units have weights that are learned during the training procedure.

The first step in LSTM is through a sigmoid layer, so called the *Forget Gate*. As shown in (1), input values are  $h_{t-1}$  and  $x_t$ , and the output is a number between 0 and 1. And  $W$  is the parameter matrices,  $b$  is the bias vector. 1 represents "completely keep this" while a 0 represents "completely get rid of this."

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (1)$$

The next step contains two parts as shown in (2) and (3). One is a sigmoid layer called the *Input Gate* decides which values to update. The other one is a tanh layer creates a vector of new candidate values  $\hat{C}_t$ , which could be added to the state.

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (2)$$

$$\hat{C}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \quad (3)$$

Then as (4) show we update the old cell state  $C_{t-1}$  into the new cell state  $C_t$ .

$$C_t = f_t * C_{t-1} + i_t * \hat{C}_t \quad (4)$$

The final output based on cell state. As (5) and (6) show, firstly run a sigmoid layer which decides what parts of the cell state to output and secondly put the cell state through tanh and multiply it by the output of the sigmoid gate.

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (5)$$

$$h_t = o_t * \tanh(C_t) \quad (6)$$

The structure of one of the LSTM neuro consists of the above procedures is shown at Fig. 2.

### B. LSTM Model Development

We split the prepared dataset into train sets and test sets. Next, we fit the model on the first 8 month of data and evaluate it on the remaining 4 months.

After adjusting the parameters, we build the LSTM model with 50 neurons in the first hidden layer and 1 neuron in the output layer to do the prediction. The input shape is 1 time step with 8 features. And we use the Mean Absolute Error (MAE) loss function and chose the efficient Adam optimization algorithm. The model fits for 300 training epochs with a batch size of 180. The model is compiled using *TensorFlow* as backend.

### C. LSTM Forecasting and Evaluation

We use the fitted model to forecast the entire test dataset. This step is accomplished by combining the forecast with the test dataset and inverting the scaling. With forecasts and actual values in their original scale, we calculate an error score for the model. We calculate the RMSE to evaluate the accuracy.

For cooperation, we chose the ARIMA model as baseline algorithm to assess the effects of predictions.

### D. Anomaly Detection

The basis for prediction is an observed pattern and the assumption that the pattern will reoccur in the future. If this assumption does not hold true, the predicted values may be far off the measured values. That means the values are possibly abnormal data and unexplainable behavior.

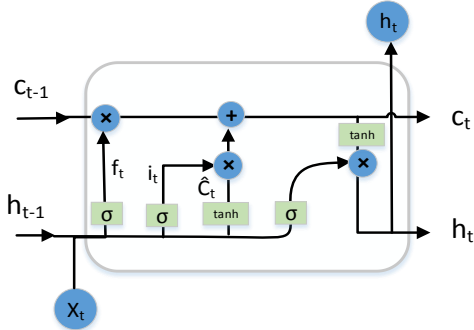


Figure 2. LSTM neuro structure

After predicting for each point in a time series, we compute the difference between predicted and observed values. The differences show the abnormality of the point in a time series. We calculate the anomaly scores normalized by the average deviation from the model to describe the possibility a point is anomaly. Furthermore, the presence of consecutive outliers means that there could be an anomaly in the sequence.

Computation of the anomaly score is showed in (7). Then we use *Min-max Normalization* to convert the score to the range of 0 to 1, the finally score  $\tilde{s}_t$  is shown in (8).

$$s_t = \frac{|predicted_t - observed_t|}{avg_{i \in T}(|predicted_i - observed_i|)} \quad (7)$$

$$\tilde{s}_t = \frac{s_t - \min(S)}{\max(S) - \min(S)} \quad (8)$$

After the anomaly scores for each point of the time series are calculated, a threshold  $\varepsilon_1$  is set to mark the abnormal point when the anomaly score is larger than  $\varepsilon_1$ .

Besides, a window size  $W$  and another threshold  $\varepsilon_2$  is set to determine if a time series is abnormal. When the number of abnormal points in the window is greater than the threshold  $\varepsilon_2$ , the consumption data series is considered as anomaly and the user is suspected of stealing electricity in some degree.

## V. EXPERIMENT RESULTS

### A. Consumption Prediction

We compare the performance of the proposed anomaly detection method with a typical time series algorithm ARIMA.

ARIMA is a generalization of the simpler *AutoRegressive Moving Average* and adds the notion of integration. The parameters of the ARIMA model include  $P$ ,  $d$  and  $q$ .  $P$  is the lag order, which means the number of lag observations included in the model.  $d$  is called the degree of differencing.  $q$  is the size of the moving average window, also called the order of moving average.

The summary of the fit model is showed as Fig. 3, which includes coefficient values used as well as the skill of the fit on the on the in-sample observations.

A plot of train and test loss over the 300 epochs of LSTM is shown in Fig. 4. The solid line represents the training set error and the dotted line represents the test set error. We can see that test loss drops below training loss in first few epochs in Fig. 4. Its means the model may be overfitting the training data. But when the number of epochs over 75, test loss turns to be on the top of training loss. The LSTM model overcomes the overfitting problem. We then measure RMSE to check if the value shed more light on this.

|                    | coef    | std err   | z       | P> z      | [0.025 | 0.975] |
|--------------------|---------|-----------|---------|-----------|--------|--------|
| const              | 11.0497 | 3.349     | 3.299   | 0.001     | 4.486  | 17.614 |
| ar.L1.D.3317000550 | 0.8213  | 0.048     | 17.001  | 0.000     | 0.727  | 0.916  |
| ar.L2.D.3317000550 | -0.1156 | 0.064     | -1.815  | 0.070     | -0.240 | 0.009  |
| ar.L3.D.3317000550 | 0.0484  | 0.064     | 0.756   | 0.450     | -0.077 | 0.174  |
| ar.L4.D.3317000550 | -0.0857 | 0.064     | -1.346  | 0.179     | -0.210 | 0.039  |
| ar.L5.D.3317000550 | 0.2765  | 0.048     | 5.727   | 0.000     | 0.182  | 0.371  |
| Roots              |         |           |         |           |        |        |
|                    | Real    | Imaginary | Modulus | Frequency |        |        |
| AR.1               | 1.0298  | -0.0000j  | 1.0298  | -0.0000   |        |        |
| AR.2               | 0.6849  | -1.0859j  | 1.2839  | -0.1604   |        |        |
| AR.3               | 0.6849  | +1.0859j  | 1.2839  | 0.1604    |        |        |
| AR.4               | -1.0449 | -1.0193j  | 1.4597  | -0.3770   |        |        |
| AR.5               | -1.0449 | +1.0193j  | 1.4597  | 0.3770    |        |        |

Figure 3. Summary of the fit ARIMA model

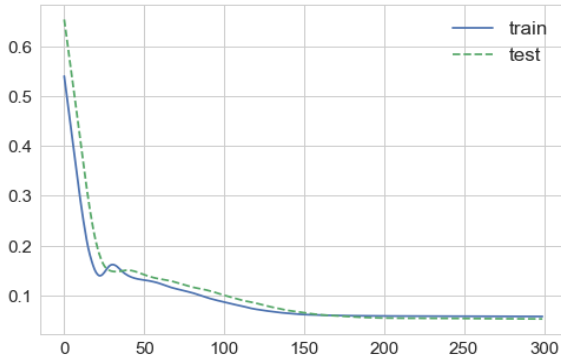


Figure 4. Train and test loss over the 300 epochs of LSTM

Fig. 5 shows the density plot of the residual error values of ARIMA, suggesting the errors are Gaussian, but not centered on zero.

Through calculation, average test RMSE of LSTM model is 3.05 while the value of ARIMA is 3.72. LSTM show 22% advantage in accuracy on this prediction. In Fig. 6, the red line represents the real consumption data and the blue line represents the prediction data. It shows that LSTM model did an excellent job of fitting the test datasets, while ARIMA missed some pattern in one typical user consumption curve. The addition of weather and calendar information makes predictions based on LSTM more robust and accurate.

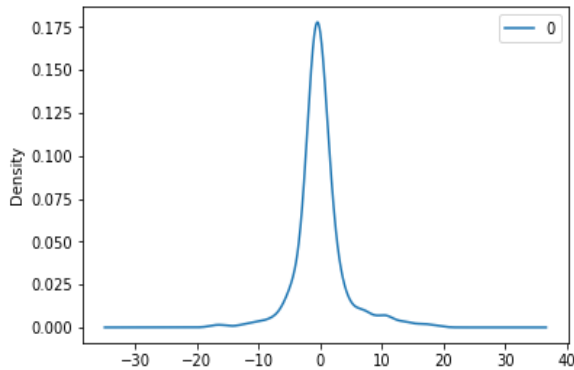


Figure 5. Residual error values of ARIMA

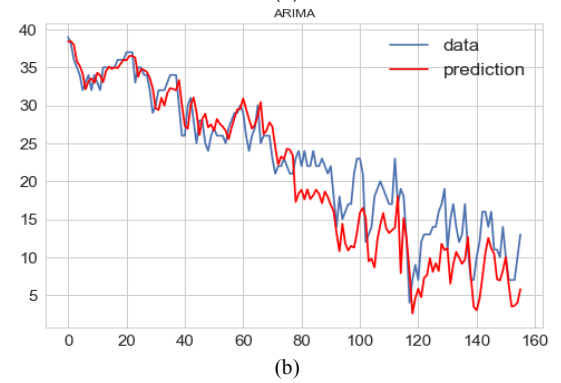
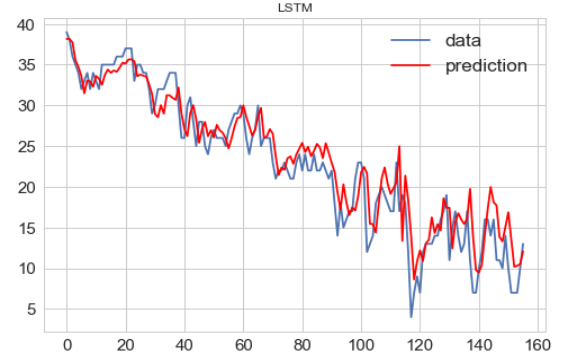


Figure 6. LSTM and ARIMA prediction comparison

## B. Anomaly Detection

After numbers of rounds of adjustment and verification, we found the appropriate threshold  $\varepsilon_1$ ,  $\varepsilon_2$  and window size  $W$  for our dataset.  $\varepsilon_1$  is set to be 0.63 to mark the single abnormal point in the time series, while the window size i.e. the detection range is 21. And the threshold  $\varepsilon_2$  is set 9, which means an anomaly in the series is found while there is over 9 single abnormal points during the consecutive 21 days.

In practice, we can adjust threshold and window size to increase or decrease the severity of detecting according to the requirements of the scene. Higher  $\varepsilon_1$  and lower  $\varepsilon_2$  can recall fewer anomalies and reduce misjudgment rate. Scores of one user's consumption curve is shown in Fig. 7. The threshold  $\varepsilon_1$  and anomaly windows are marked by red dotted lines and green dotted boxes, respectively.

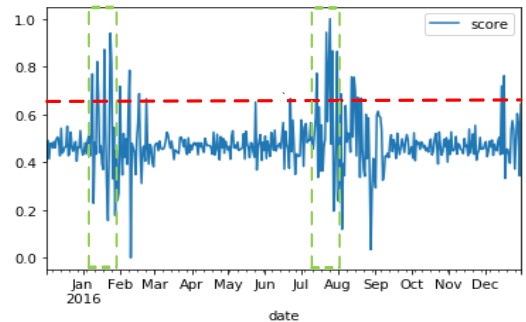


Figure 7. Scores of one user's consumption curve

One of the major challenges for evaluating anomaly detection methods is the unavailability of fine-grained ground truth data about the actual anomalies. In our original datasets, there are 396 users labeled as electricity thief by some detecting machine and field inspection verification. These labeled data help us to verify the ability of LSTM model in identifying some known anomalies patterns.

As a comparison, we also use clustering algorithm to find power stealing users. K-means algorithm is one of the most widely applied clustering algorithms. K-means clustering is an iterative algorithm. First, randomly select  $k$  random point as clustering center. Second, for each data point in the data set, it is associated with the nearest center point according to the distance from  $K$  center points. All points associated with the same center point are clustered into a class. Third, calculate the mean value of each group and move the center point to the position of the mean value of the group. At last, repeat the last 2 steps until the center point does not change or meet the stop condition. Users with abnormal fluctuation in the normalized electricity consumption data form isolated clusters or aggregate into classes far from other clustering classes. Using K-means algorithm we found 125 abnormal users who are suspect of stealing power, of which 103 users in the electricity thief list of source datasets. The accuracy is 0.82 and the recall rate is 0.26.

The LSTM model marked 9,784 anomaly points of 271 users. There are 601 anomaly windows pass the threshold  $\epsilon_2$ , which come from 231 users. 134 users of the total 231 users are in the electricity thief list of source datasets. The accuracy is 0.71 and the recall rate is 0.58. Our LSTM model highly improves the recall rate.

Focusing on unlabeled abnormal users that our LSTM model identified, three domain experts help us to analyze them to determine whether the abnormal data is real or not based on their rich experience combined with domain knowledge. 24 users are concerned as abnormal of which the signatures are unknown anomalies patterns like unexpected drops, spikes and trend changes that is different from the labeled theft users.

## VI. CONCLUSION

Prediction of power consumption data and identification of abnormal power consumption data are the key and tough points to ensure the operation efficiency and reliability of smart grid. We propose a power consumption forecasting and anomaly detection model based on LSTM, which shows an accuracy advantage comparing to other time series prediction algorithm. The detection result is further used to identify the anomaly and electricity theft which shows better practicability than unsupervised algorithm. Model construction and validation are based on real user datasets. In

future, we will further optimization algorithm to increase the prediction and anomaly detection accuracy.

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